

Yafira Martinez

Design Engineer / Creative Technologist

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Interactive Systems, Interfaces & Experimental Technologies

SKILLS

Front-End Development: React, Next.js, TypeScript, JavaScript, HTML/CSS, Node.js, REST APIs, Web Serial, p5.js, ml5.js, Creative Coding, Supabase, Vercel, Git

Design: Interaction Design, UX/UI, Responsive Web Design, Design Systems, Prototyping, GenAI Tools, Wireframing, Digital Accessibility, Information Architecture, User Research, Figma, Adobe Suite

Hardware: Physical Computing, Microcontrollers (Arduino, Raspberry Pi, ESP32, RP2040, Adafruit), CircuitPython, Python, Fritzing, E-Textiles, Soft Circuits, Wearables, Soldering, Fabrication (laser cutting, 3D printing, knitting machine)

EXPERIENCE

Design Engineer & Creative Technologist — Electrocute Lab

Mar 2021 – Present

Creative independent practice

- Design and ship independent web applications and tools end to end — electrocute-ui, a React component library with an 18-color design token system published to npm, and tinytinker.tools, an open-source maker toolkit of 22+ browser tools — alongside maker tutorials, open workshop materials, and project documentation on a technical blog (Astro + TinaCMS).
- Work at the intersection of textile craft and computation — soft circuits, e-textiles, physical computing, and generative text — with projects exhibited at venues including the Open Hardware Summit in Berlin and taught in public workshops.

Lead Camp Counselor — ITP Camp, New York University

Jun 2026

Previously: Camp Counselor, Jun 2025

- Led a 150+ person creative technology community through workshops, mentorship, and hands-on support across hardware, software, and fabrication.
- Designed and built spot-mapper, a real-time collaborative floor-plan mapping tool (Leaflet, Vercel serverless functions, Redis, token-based auth) used live during the Show All Things Show, with admin/viewer roles, mobile support, and full handoff documentation.

Hardware Research Developer — NYU Ability Lab

Oct 2025 – May 2026

- Researched accessible fashion technologies for the Making Fashion Accessible project, using knitting machines and AYAB (open-source knitting machine software) to support inclusive, open-source garment production and adaptable textile workflows.

Graduate Teaching & Research Assistant — NYU ITP/IMA

Sep 2025 – May 2026

- Graduate Teaching Assistant for Creative Computing (Prof. Daniel Shiffman), mentoring students in creative coding and generative systems.
- Graduate Teaching Assistant for Intro to Wearables (Instructor Daniel Ryan Johnston), mentoring students in wearable design and physical computing.
- Research Assistant for Computational Text (Prof. Daniel Shiffman), developing syllabus material spanning procedural text, data ethics, and machine learning.

Machine Learning Researcher & Web Contributor — ml5.js

Jun 2025 – May 2026

Mentors: Daniel Shiffman & J.H. Moon

- Researched and prototyped ml5.transcribe() and ml5.embeddings(), bringing browser-based transformer models (Whisper, Moonshine) to a library designed for students, artists, and beginner coders.
- Refined interaction design and inference performance; contributed documentation and examples supporting both contributors and learners.

Web Developer & Designer — Useless Machines Archive, NYU

Mar 2025 – Aug 2025

Professor: Blair Simmons

- Designed and built a front-end web archive using the Are.na API, focusing on client-side data fetching, content organization, and responsive interface design.

Lead Web Developer — Equitable Syllabus Project, NYU ITP/IMA

Nov 2024 – May 2026

- Redesigned and rebuilt the project website with accessible, user-centered interfaces — from Figma designs to a production Next.js deployment — while migrating project data and maintaining centralized contribution tracking.

Software Engineer & Creative Technologist Intern — Floreo Labs

Jan 2020 – May 2020

- Developed responsive front-end applications in React and designed UI/UX prototypes in Adobe XD; contributed to mobile projects while managing agile workflows and producing technical documentation.

EDUCATION & CERTIFICATIONS

Master of Professional Studies (MPS), Interactive Telecommunications Program

May 2026

New York University, Tisch School of the Arts

Bachelor of Science (BS), Computer Science

Dec 2020

City University of New York, Lehman College

Certifications: Artificial Intelligence, Samsung Innovation Campus (2022–2023) · Human Computer Interaction, Interaction Design Foundation (2022) · Google UX Design, Coursera (2021)

SELECTED WORK & EXPLORATIONS

computer.garden() — Processing Community Day @ NYC

Oct 2026

Generative sketch that grows soft-circuit gardens using breadth-first search (BFS) maze-routing, rendered in a Bayer-dithered e-ink aesthetic.

Tiny Soft Computers Workshop — SOFTER Digital Futures (Remote Residency + Conference), Denmark

Aug – Oct 2026

Co-developed an intro to physical computing and alternative interfaces with Alanna Okun in a remote residency, then co-taught it in person at the conference to 16 participants. Led hardware and software (e-ink kit, CircuitPython) and built the workshop site, which doubles as open-source documentation for educators.

Intro to Machine Learning Workshop — NYU College & Career Lab, NYC

Jul 2026

Designed and co-led a hands-on workshop for ~60 middle school students, building a Teachable Machine and ml5.js + p5.js image-classification game to introduce students to training and using their own ML models.

Soft Circuit Boards (SCBs) Workshop — ITP Camp, NYU

Jun 2026

Designed and taught a hands-on workshop on handmade circuits — copper tape traces, craft foam boards, and LEDs, no soldering required — extending work from the Soft Circuit Library.

PetalByte — Bring Your Own Data (BYOD), NY Tech Week by a16z, NYC

Jun 2026

Portable generative text device treating the USB drive as both archive and interface; Markov chain corpora live on a flower-shaped laser-cut acrylic enclosure, browsable through a custom pygame interface on a Raspberry Pi touchscreen.

The Soft Computer Mini — Open Hardware Summit, Berlin

May 2026

Travel-ready extension of The Soft Computer, exploring embodied input, generative text, and textile as substrate, compressed into a tarot-sized device.

The Soft Computer — Periphery: ITP Thesis Show, NYC Resistor

May 2026

Textile-based computing interface combining machine-knit wool, capacitive fabric inputs, haptics, and a flexible e-ink display; Raspberry Pi 5 drives Markov chain text generation across four corpus-mapped channels.

Open Hardware Summit Fellow — Open Source Hardware Association, Berlin

Feb 2026 – Jun 2026

Selected fellow exhibiting The Soft Computer and other experimental open-source hardware projects at the Open Hardware Summit, Berlin.

Soft Circuit Library — Electronics Faire (Theme: Repair), Temple University Libraries

Apr 2026

Collection of handmade textile circuit boards organized as a living archive, each designed to be handled, studied, and repaired by touch.

Guardians of the Groceries — Best of Software, PlayfulNYC Hackathon

Apr 2026

Awarded Best of Software. Built the hardware and software layer of a handheld ESP32 device that captures photos, uploads to Supabase via WiFi, and pairs with a live web registry documenting NYC bodega cats.

Pocket Oracle of Time — Horological Society of New York

Feb – Apr 2026

Poetic timekeeping object exploring ritual, slowness, and material interaction with time through embedded electronics and tactile design.

Enoki — MIT Media Lab AI x Hardware Hackathon

Mar 2026

Focus companion for communal accountability: a React Native app using Mentra smart glasses and Claude Vision AI paired with a Raspberry Pi desktop device that visualizes focus as a mushroom that grows when on task and wilts when distracted.

Moon Pocket — Pedestal Playtime: Art Toy & Tech Toy Open Mic, NYC Resistor

Jan 2026

Connected device (Arduino Nano 33 IoT) that fetches real-time moon and tide data from public web APIs and renders it on a round TFT display through touch-based interaction.

Tendertronics — NYU ITP NIME Show, Littlefield

Dec 2025

Analog instrument performance exploring tactile, minimal interfaces and embodied sound interaction through handcrafted circuitry.

Lightning Talk Speaker — Exploring Creativity in the Age of AI, NYU Global Show & Tell

Oct 2025

Presented "Open Tools, Soft Interfaces" on ml5.js contributions and framing creativity in AI through openness, design as dialogue, and soft, human-centered interfaces.

Static Chic — Volt Couture: Wearables + E-textiles Open Mic, NYC Resistor

May 2025

3D-printed inductive handbag with an embedded coil that wirelessly powers and lights up LEDs, presented at an open mic curated by Kari Love and Liza Stark.